precept-research: architecture decisions, methodological risk and forward tickets

Gian Prem Rajaram · UCL MSc CS · 25 July 2026 · Companion to the ground-up status document

Evidence-backed design review. Every recommendation carries its counter-case. Section 9 is written in the repo's DSE ticket format so it can be handed to Claude Code directly. Sources listed in section 10.

1. Bottom line up front

The field moved under you between the lit review and today, and it moved in a direction that is mostly good news. Three things changed. (1) A benchmark now exists that fixes the exact defect that makes Who&When a poor CPVI substrate: TraceElephant (ACL 2026) records each agent's *input context* as well as its output, which is precisely the conditioning state s that your P0-1 fix made mandatory - and Who&When does not record it. (2) The 14.2 per cent step-accuracy headroom your lit review leans on is stale; AgenTracer-8B and TraceElephant's own agentic methods now sit around 20 to 33 per cent, so the H5 claim needs re-anchoring or it will read as a straw man at viva. (3) AgenTracer solved your Y-on-logs problem in a citable way, with counterfactual replay, and TraceElephant ships the runnable environments and a non-intrusive API middleware that make replay and live gating possible on a real multi-agent system rather than only on your simulator.

Four decisions follow, in descending order of how much they change the dissertation. (1) Move RQ3a's primary substrate to TraceElephant and demote Who&When to a comparability anchor. (2) Define Y on logs by counterfactual replay rather than by annotation, which removes the circularity and is now precedented. (3) Cut SocialJax formally, on evidence rather than schedule - it has no communication channel at all, so the comparison would require building one, training comm-capable MARL, and redefining CPVI over continuous vectors. (4) Harden the probe methodology with control-task selectivity, which is the standard defence against 'your probe learned the task, not the representation' and which your current design does not have.

What I would not change. The simulator, the channel ladder, the receiver-conditioned CPVI construct, the calibrate-against-realised-failure discipline and the matched-control gate design are all sound and, in the case of the calibration circularity guard, better than what the surrounding literature does. The recommendations below are additive; none of them require unpicking RQ1 or RQ2.

2. How the field moved, and what it costs you

Signals: five benchmarks now occupy the failure-attribution space your RQ3a targets, three of them published after the 25 July lit review was drafted. The table is ordered by relevance to CPVI. Abbreviations: MAS = multi-agent system; GT = ground truth (whether the attribution method is told the correct final answer); alpha = Krippendorff's alpha; kappa = Cohen's kappa.

Benchmark	Size	Licence / access	Records what	Why it matters to you
TraceElephant (ACL 2026)	380 executions, 220 annotated failures	CC-BY-4.0; GitHub + HF TraceElephant/ TraceElephant	Full traces: input_context AND output_content per step, plus tool logs, agent config, system architecture. Runnable environments shipped.	The only substrate whose schema maps onto HandoffRecord without approximation: input_context is your observation (the conditioning s), output_content is your message m. Also 160 non-failure executions, so trace-level Y is genuinely two-class. Also enables counterfactual replay and live interception.
Who&When (ICML 2025)	184 annotated failure tasks	MIT; HF Kevin355/ Who_and_When	Agent outputs only. Inputs and system-constructed context are not recorded.	The published baselines everyone knows, so it stays as the comparability anchor. But all 184 are failures (single-class Y) and s is not recoverable - the two defects that make your RQ3a hard.
TracerTraj / AgenTracer (2025)	2,000+ annotated traces	See paper; AgenTracer-8B released	Failed trajectories annotated by counterfactual replay and programmed fault injection across 6 frameworks and 6 benchmarks.	The method, not the data, is the prize: counterfactual replay is a non-circular way to define a per-step outcome. This is the citable precedent for your Y-on-logs fix.

MAST-Data / MAD (NeurIPS 2025 D&B)	1,642 traces, 7 frameworks	CC-BY-4.0; HF mcemri/ MAST-Data	Trace-level 14-mode binary failure annotations (3 categories), taxonomy built from 150 traces at kappa 0.88.	Trace-level only, no step localisation, so it cannot test H5's localisation claim. Useful as a secondary corpus for the coarse 'does low CPVI coincide with inter-agent-misalignment traces' question. The HF preview shows all-zero annotation rows, so a no-failure-mode class appears to exist - verify the proportion.
TRAIL (2025) and TELBench (2026)	148 traces / 841 errors; 2,790 trajectories, 1,000-instance benchmark	Public on HF	TRAIL: taxonomy-annotated GAIA and SWE-Bench traces. TELBench: span-level harmful-error annotations in deep-research trajectories.	Both are largely single-agent scaffolds, so there is no inter-agent handoff to score. Cite as scope-adjacent, do not build loaders.

2.1 The stale-headroom problem, and what to write instead

Signals: your lit review positions H5 against 53.5 per cent agent and 14.2 per cent step accuracy. Both numbers are from the original Who&When paper and both have been beaten. Writing to the old numbers is the single most likely thing to draw a hostile viva question.

Method	Best reported	How to treat it
Who&When published baselines (all-at-once, binary search, step-by-step)	53.5% agent, 14.2% step	Keep as the historical anchor and as the floor. State the date.
AgenTracer-8B (2025)	Beats Gemini-2.5-Pro and Claude-4-Sonnet by up to 18.18% on Who&When; reported around 69% agent and 20% step	This is the number a reviewer will know. Your H5 must clear or explain itself against it, not against 14.2%.
TraceElephant static/dynamic agentic (2026), full traces	66.7% agent, 33.3% step with GT; 60.6% / 27.6% without GT	If you move substrate, these are your baselines. They are strong, which changes the framing of the contribution.
Frontier reasoning LLMs, generic	Generally below 10% on step attribution	Still the honest statement of how hard the task is in general. Safe to keep.

Reframing that survives the change. Do not claim CPVI beats a specialised, RL-trained failure tracer at localisation. It probably will not, and it is not what the construct is for. The defensible claim is narrower and better: CPVI is a *cheap, prospective, per-handoff* score computable at the moment of the handoff, whereas every method above is a retrospective analysis of a completed failed trace with the outcome already known. AgenTracer-8B is an 8-billion-parameter model reading a whole trajectory after the fact; s_cos is a cosine. The comparison to make is on the operating characteristic - what localisation you get per unit of compute, and whether you get any of it before the outcome exists - not on raw accuracy. That framing is both more honest and more novel, and it is the one that connects RQ3a back to RQ3b, where a retrospective tracer is useless by construction.

3. Decision 1: the RQ3a substrate

3.1 Why Who&When transferability is worse than it looks

Three distinct problems, which the current methodology text treats as one. Separating them is what makes the decision tractable.

Problem A: the conditioning state is not in the data. CPVI is defined as the log-loss reduction of a probe on [s ; m] over a probe on [s], where your own pre-registered semantics fix s as the state observable to the receiver. Who&When records agent outputs. It does not record the input context each agent received - the system-constructed prompt, the retrieved documents, the orchestrator's framing. TraceElephant's authors demonstrate this directly: they took a Who&When failure case, re-ran the original system to restore the missing inputs, and showed the failure cause is ambiguous under partial observability and largely resolved with full inputs. They quantify it at 22 per cent agent-level and 76 per cent step-level accuracy degradation when inputs are removed. For you the consequence is sharper than for them: you would have to *reconstruct* s by concatenating preceding outputs, which is a different quantity from what the receiver actually saw, and the entire conditional construct rests on s being right. A reviewer who has read TraceElephant will ask this question.

Problem B: the outcome is single-class. All 184 Who&When instances are failures. A probe cannot be fitted on a constant label, so the 'refit probes on a held-out portion of the logs' arm in your methodology is not merely difficult, it is undefined as written. This is the hole the first document flagged; it is worth restating that it is a property of the corpus, not of your method.

Problem C: the domain gap is large, and it is a probe-transfer question nobody has answered. Your simulator probe is fitted on bge-base embeddings of numeric, grid or natural-language physics state paired with terse movement instructions. Who&When is GAIA and AssistantBench web research and tool use. The encoder is shared, so the vectors live in one space, but a logistic probe is a hyperplane fitted to one region of that space; there is no reason its decision boundary is meaningful in another. The V-information literature has no established transfer result - Ethayarajh et al. apply PVI within a dataset, Hewitt et al. within a corpus, Lu et al. in-context on the same task family. You would be making a transfer claim the construct's own literature has not made. That is not a reason to avoid it. It is a reason to report it as an explicit, framed experiment with an expected-null branch, rather than as a routine second arm.

3.2 The options, evaluated

Signals: four substrate strategies. Cost is in your working days. Risk is the probability the arm produces an uninterpretable result rather than a reportable one.

Option	What you do	Upside	Cost / risk	Verdict
A. Stay on Who&When	Reconstruct s from preceding outputs; use the transfer arm only, since refit is unfittable.	Zero re-planning. The published baselines are directly comparable. Reviewers know the benchmark.	3-5 days. High risk: s is an approximation, so a null is uninterpretable (bad construct or genuine absence?).	Not as primary. Keep as anchor.
B. Move primary to TraceElephant	Map input_context to observation and output_content to message; refit on the mixed corpus; keep Who&When for baseline comparability.	Schema maps to HandoffRecord with no approximation. 160 non-failures give a fittable two-class Y. Runnable environments enable replay and live gating. CC-BY-4.0.	5-8 days. Medium risk: newer benchmark, smaller community, you must justify the switch in the methodology.	Recommended primary.
C. Add MAST-Data as secondary	Trace-level coarse test: do low-CPVI traces carry the inter-agent-misalignment labels?	1,642 traces, largest corpus, strong provenance (kappa 0.88), CC-BY-4.0.	2-3 days. Low risk but low reward: no step localisation, so it cannot test H5 as stated.	Yes, as a cheap secondary.
D. Generate your own corpus	Use TraceElephant's LLM API middleware to instrument a real MAS and collect fresh traces with full inputs and known outcomes.	Total control of s, Y and success balance. Directly enables RQ3b on a real system.	8-12 days plus API cost. Medium-high risk of scope creep.	Stretch only, and only if RQ1 lands early.

Recommendation: B primary, A as anchor, C as cheap secondary, D as a stretch flagged in future work. Concretely: report H5 on TraceElephant with both refit and transfer arms, then report the transfer arm alone on Who&When purely so the numbers sit next to the published baselines. Say in the methodology exactly why - that Who&When's output-only traces do not contain the receiver's input context on which the conditional construct depends, citing TraceElephant's own quantification of that gap. Written that way, the substrate switch stops being a retreat and becomes a methodological argument in your favour: you are the only person in this space who *needs* full observability, because you are the only one conditioning on it.

The counter-case, stated fairly. Who&When is the ICML spotlight with 367 GitHub stars and the numbers everyone cites; TraceElephant is a 2026 ACL paper with 11 stars. If your examiners are not close to this literature, switching substrate looks like moving the goalposts. Two mitigations: keep the Who&When transfer arm so the familiar numbers appear in the table, and make the observability argument in one crisp paragraph early in the RQ3a chapter rather than in a footnote.

4. Decision 2: what Y is on real logs

Signals: this is the deepest methodological risk in the project, because it is the one place where a wrong choice produces a result that looks fine and means nothing. Four candidate definitions; the third is now precedented and is the recommendation.

Definition	Construction	Circularity status	Verdict
Y1 Trace success	Y = did the whole task succeed. Requires a mixed corpus; available on TraceElephant (160 of 380) and possibly MAST.	Clean. The label comes from task outcome, not from the attribution annotation.	Use as the primary refit label. Weak per-handoff signal (long credit path), which you already know from y_terminal_success.
Y2 Annotated decisive step	Y = is this the decisive error step, per the human annotation.	Circular. You would train the probe on the label the localisation claim is evaluated against.	Forbidden. Name it explicitly in the methodology as considered and rejected - that sentence does real work at viva.

Y3 Counterfactual replay	Re-run the system from step t with the step's output corrected or substituted; Y = did the outcome change. Requires runnable environments, which TraceElephant ships.	Clean, and structurally identical to the definition of a decisive step without using the annotation.	Recommended where compute allows. Precedented by AgenTracer, which built TracerTraj this way, and by Jaques et al.'s counterfactual causal-influence reward.
Y4 Annotation-free local proxy	Y = is the receiver's next action schema-valid / does it advance the recorded plan.	Clean if and only if the proxy never references the annotation.	Cheap fallback when replay is too costly. The proxy design needs its own audit, so budget for that.

Why counterfactual replay is the right answer and not just a clever one. Your whole thesis rests on the Lowe et al. argument that a correlation between a message statistic and an outcome is not evidence the message mattered, and that only intervention settles it. Counterfactual replay is that same intervention applied to the label rather than to the gate. Defining Y by replay means the outcome variable is itself interventional, which makes the RQ3a localisation claim and the RQ3b causal claim rest on one consistent epistemology instead of two. AgenTracer built a 2,000-trace dataset this way and trained a model that beats frontier LLMs on the task, so the technique is validated at scale. Jaques et al. established the same counterfactual logic for measuring influence in MARL seven years earlier. You would be the first to use it to define the *target* of an information-theoretic boundary measure, which is a genuine methodological contribution worth a paragraph in its own right.

The honest cost. Replay means re-running real agent systems, which means API spend and wall time, and it means non-determinism in the replay itself (the systems are LLM-driven). Two mitigations, both pre-registrable: replay each counterfactual n times and define Y by majority outcome, reporting the replay agreement rate as a data-quality statistic; and restrict replay to a stratified sample of steps rather than every step, since the probe does not need every handoff labelled. Budget this as a bounded experiment with a fixed spend cap, not an open loop.

5. Decision 3: SocialJax, and why the cut is evidenced rather than scheduled

You asked why this is being skipped. The scheduling argument in the roadmap is the weak version. Here is the real one.

SocialJax has no communication channel. The suite (ICLR 2026, Apache-2.0) ships Coins, Clean Up, Commons Harvest in three variants, Territory, Coop Mining, Mushrooms, Gift Refinement and PD Arena - all gridworld sequential social dilemmas derived from Melting Pot 2.0, where agents interact through spatial actions and reward structure, not messages. The shipped algorithms are IPPO, MAPPO, VDN, SVO (social value orientation reward shaping) and TRANSFER (self-interest reward exchange). None of them is a communication algorithm. There is no message to score, and therefore nothing for CPVI to measure.

What the arm would actually cost, itemised:

Work item	Why it is needed	Realistic cost and risk
Add a message channel to a SocialJax environment	There is no channel. You would extend the observation and action spaces and re-verify the Schelling-diagram social-dilemma property still holds.	3-5 days, and it invalidates the suite's own validation of the environment unless you redo it.
Implement a comm-capable MARL algorithm	IPPO/MAPPO/VDN do not learn messages. You would port something like CommNet, TarMAC or IMAC into the JAX stack.	5-10 days. This is the part that most often silently fails.
Train to convergence on GPU	Learned communication needs training, unlike your frozen-LLM setting.	GPU-days, competing directly with the RQ1 sweep for the same Myriad allocation.
Redefine CPVI over continuous message vectors	Your V-information argument is built on the dimensionality problem for text embeddings and a bounded probe family. A learned 16-dim message vector is a different estimation regime; PVI is well defined but the motivating argument is not the same one.	2-3 days of methodology writing, and it weakens rather than strengthens the construct argument.
Interpret a cross-paradigm comparison	Frozen text messages scored versus learned vector messages trained end to end. Any difference is confounded by the training regime, the message space and the task.	Uninterpretable without a much larger design. This is the killer.

Recommendation: cut, and say why in the limitations rather than in the schedule. The sentence to write is that the learned-message setting is a different measurement regime, not a harder version of the same one: IMAC (Wang et al. 2020) and the emergent-communication line optimise a message encoder end to end and presuppose gradient access, whereas this dissertation scores arbitrary frozen natural language at inference with no access to the sender. A comparison between them would confound the training regime with the message space. That is a stronger limitation paragraph than 'we ran out of time', and it is true.

The cheap replacement, if you want the contrast at all. Run the Eccles absent-versus-unused decomposition on your *own* data instead. You already have everything needed: a message that carries low CPVI is an absent signal; a message that carries high CPVI whose receiver still acts wrongly is an unused signal. Splitting your C0-C4 handoffs on that two-by-two costs a day of analysis, uses no new compute, cites the same literature the SocialJax arm was meant to engage, and converts a possible RQ1 null into a reportable finding about which half of the channel failed. This is strictly better value than SocialJax for a fraction of the cost, and it should be pre-registered before the main sweep.

6. Decision 4: probe methodology hardening

Signals: three additions, all cheap, all standard in the probing literature, none currently in your design. The first is the one an examiner from an NLP background will ask about by name.

6.1 Control tasks and selectivity (Hewitt and Liang, EMNLP 2019)

The problem their paper names is exactly yours. A probe that predicts well might be reading the representation, or it might simply be a capable model that learned the task. Their fix is a control task: pair each input with a random output that *by construction* can only be learned by memorisation, then report selectivity, the gap between real-task accuracy and control-task accuracy. A probe with high accuracy and high control accuracy is telling you about the probe, not the representation. In your setting the control task is easy to build: assign each handoff a random binary label stratified to match the Y base rate, refit g_cond and g_base, and report the CPVI you get. It should be approximately zero. This is a different and stronger check than your existing shuffled-message audit, which permutes messages within condition and tests whether the message carries signal; the control task tests whether the *probe family itself* can manufacture apparent information from noise at your sample size. With 1,536-dimensional concatenated features and a few hundred pilot handoffs, that is a live risk, and auroc_train_cond alone does not settle it.

6.2 Repeated cross-fits, because per-handoff scores are noisy

A single K-fold cross-fit gives each handoff its score from one fold model, so per-instance signs can flip under probe reseeding. The scores that feed your mediation and all of RQ2 should be averaged over R repeated cross-fits with different fold seeds, with the across-repeat standard deviation reported as the score's measurement error. sklearn logistic refits are milliseconds at this scale, so R = 5 to 10 is nearly free. Pre-register R. Without this, the H2 indirect effect inherits probe-fit noise that nothing in the current design quantifies.

6.3 Length control, because it is the first attack line

'CPVI is a fancy word count' is the obvious objection and C1 makes it structural, since C1 manipulates message length directly. Pre-register a partial Spearman of CPVI with outcome given token length, and length as a covariate in path b of the mediation. Done properly this inverts the attack: C1 becomes the cell that demonstrates the construct and the confound dissociate. The emergent-communication literature has the same problem with channel capacity as a confound and handles it with matched ablations, so the move is well precedented.

7. What this means for the codebase

Signals: the recommendations are additive. Nothing above requires unpicking the simulator, the channel, the estimator or the gate. Five touch points, with the design constraint each one has to respect.

Component	Change	Design constraint it must respect
data/schema.py	No change to HandoffRecord. Add a separate LogHandoffRecord for real-log rows, or reuse HandoffRecord with a source discriminator.	Do not widen the simulator schema to accommodate logs. The RunManifest and ExperimentConfig schemas are your stable contract; a log-side record is a different contract and should be versioned separately. Reusing HandoffRecord with nullable physics fields would weaken the type discipline that makes the simulator side auditable.
measure/featuriser.py	Unchanged. It already embeds observation, not state_str, so a log row with input_context in the observation slot flows through untouched.	The content-addressed cache is keyed on encoder name, revision and text, so simulator and log embeddings can share one cache safely. Do not add a domain field to the cache key; the vectors are genuinely the same function of text.
measure/pvi_cpvi.py	Add the control-task path and repeated cross-fits. Both are new functions, not modifications to cpvi().	Keep StratifiedGroupKFold grouping on episode_id; on logs the group key is trace_id. The leakage discipline is the same discipline, and the group key is the only thing that changes.
gate/	Unchanged for RQ3b on the simulator. If you take the stretch, RuntimeGate would sit behind TraceElephant's LLM API middleware.	Statistic and RuntimeGate are already cleanly separable, which is what makes the middleware path a new adapter rather than a rewrite. Preserve that.
experiments/rq3a*.py	New. Loader, schema mapping, replay-based labeller, localisation analysis with both regimes.	The replay labeller must be a separate module from the loader, because replay costs money and the loader must stay runnable offline for tests.

8. Updated risk register

Signals: three risks the first document did not carry, all surfaced by the research above. Each is stated with the trigger you would actually observe.

Risk	Trigger you would observe	Pre-agreed response
R7 Stale baseline framing	A reviewer or examiner cites AgenTracer-8B or TraceElephant and asks why your headroom argument uses 14.2 per cent.	Re-anchor H5 now, before writing the RQ3a chapter. Frame the contribution on the prospective-versus-retrospective axis and on cost, not on raw accuracy. Ticket DSE-047.
R8 Probe selectivity	auroc_train_cond high, held-out AUROC moderate, and a control-task CPVI that is not near zero.	Report selectivity alongside every CPVI figure. If the control-task CPVI is materially above zero, reduce probe capacity (raise regularisation, drop to the linear probe) before the freeze, and report the selection rule.
R9 Replay non-determinism	The same counterfactual replay yields different outcomes across runs, so Y is unstable.	Majority-vote Y over n replays; report the replay agreement rate as a data-quality statistic and exclude steps below an agreement floor. Pre-register n and the floor.

9. Forward tickets

Written in the repo's DSE format so they can be handed to Claude Code directly. Dependencies reference existing tickets. None of these are code changes made in this session.

DSE-041 — RQ3a substrate migration: TraceElephant loader and schema mapping

Epic: E7 · Type: feat · Priority: P0 · Effort: L (8-10h) · Phase: 6 · Dependencies: DSE-004, DSE-038 (the Y-on-logs design note)

Context. TraceElephant (ACL 2026, CC-BY-4.0) records each step as (step_id, agent_id, agent_name, input_context, output_content, tool_logs) plus trace metadata, and ships 380 executions of which 220 are annotated failures. input_context is the receiver-observed state s that CPVI conditions on; Who&When records outputs only and therefore cannot supply s without reconstruction. This ticket makes TraceElephant the primary RQ3a substrate.

Acceptance criteria. src/preceptx/experiments/rq3a_load.py loads TraceElephant from HuggingFace (TraceElephant/TraceElephant) behind the existing data extra, normalising each step into a LogHandoffRecord with observation = input_context, message = output_content, and the native annotations (responsible agent, decisive step) attached but never used as a feature.

A LogHandoffRecord Pydantic model with extra='forbid' and its own SCHEMA_VERSION, separate from HandoffRecord. Physics fields are absent, not nullable.

Inter-agent handoff extraction: identify consecutive step pairs where the acting component changes, and mark which steps are handoffs versus intra-agent tool turns.

A Who&When loader retained behind the same interface, with an explicit reconstructed_observation flag on every row it emits, so downstream analysis can never silently mix approximated and true s.

docs/rq3a_schema_mapping.md documenting the field-by-field mapping for both corpora and stating the observability caveat.

Counts reported per corpus: traces, steps, extracted handoffs, failures, non-failures.

Technical notes. Do not widen HandoffRecord. The simulator schema is the stable reproducibility contract; a log record is a different contract.

Grouping key for cross-fitting is trace_id on logs, the analogue of episode_id.

Testing requirements. Unit: both loaders parse committed fixture samples into valid rows; handoff extraction on a hand-built fixture yields the expected pairs; the reconstructed_observation flag is set only by the Who&When path.

Property: every emitted row validates against LogHandoffRecord.

Out of scope. The localisation analysis (DSE-024, to be re-scoped) and the replay labeller (DSE-042).

Definition of done. Loaders merged with tests; counts table produced; schema mapping doc committed.

DSE-042 — Counterfactual-replay outcome labeller for real logs

Epic: E7 · Type: research · Priority: P0 · Effort: L (10-12h) · Phase: 6 · Dependencies: DSE-041

Context. CPVI needs a per-handoff Y that is not the attribution annotation, or the localisation claim is circular. Counterfactual replay defines Y interventionally: re-run the system from step t with the step's output substituted, and record whether the outcome changes. AgenTracer built TracerTraj this way; TraceElephant ships the runnable environments that make it possible. This is the methodological core of RQ3a.

Acceptance criteria. src/preceptx/experiments/rq3a_replay.py exposes label_by_replay(trace, step_ids, n_replays, budget) returning a per-step Y plus a replay agreement rate.

Y is the majority outcome over n replays; steps below a configurable agreement floor are flagged, not silently dropped.

A hard spend cap and a dry-run mode that reports the projected call count and cost before executing.

Stratified step sampling so the labeller can run on a subset without biasing the failure/non-failure balance; the sampling rule is recorded in the manifest.

Trace-success Y (the cheap Y1 label) computed for every trace regardless, so the refit arm has a fallback if replay is cut.

Technical notes. Replay is non-deterministic because the systems are LLM-driven. Determinism here means low-variance and agreement-reported, exactly as for the simulator.

The labeller must not import the annotation fields at all; enforce with a signature test, mirroring the no-Y discipline already used in measure/twin.py.

Testing requirements. Unit: majority voting and the agreement floor on synthetic replay outcomes; the dry-run projection; the signature guard proving the labeller cannot read annotations.

Integration: a stubbed replay backend produces labels end to end on a fixture trace.

Out of scope. Running replay at full scale; that is a budgeted experiment, not a code deliverable.

Definition of done. Labeller merged with tests; dry-run cost projection produced for the full TraceElephant corpus.

DSE-043 — Control tasks and probe selectivity

Epic: E4 · Type: feat · Priority: P0 · Effort: M (5-6h) · Phase: 2 · Dependencies: DSE-014

Context. Hewitt and Liang (EMNLP 2019) show that probe accuracy alone cannot distinguish 'the representation encodes this' from 'the probe learned the task'. Their remedy is a control task with random labels and a reported selectivity gap. At 1,536-dimensional features and pilot-scale N this is a live risk, and auroc_train_cond does not address it. This must land before the Y/V freeze.

Acceptance criteria. measure/pvi_cpvi.py gains control_task_cpvi(e_s, e_m, y, groups, cfg, seed) which refits g_cond and g_base against random labels stratified to the observed Y base rate and returns the resulting CPVI.

CpviResult gains selectivity fields: control_mean_cpvi and selectivity = mean_cpvi minus control_mean_cpvi.

analyse_rq1 reports selectivity alongside every CPVI summary; the RQ1 figure captions carry it.

ANALYSIS_PROTOCOL gains an entry describing the check and the pre-registered expectation that control CPVI is approximately zero.

Technical notes. This is distinct from shuffled_message_cpvi: that permutes real messages to test whether the message carries signal; this randomises the label to test whether the probe family can manufacture information at this sample size. Both are needed and they answer different objections.

Reuse the same cross-fit splitter so the comparison is like for like.

Testing requirements. Unit: control CPVI is approximately zero on the informative synthetic fixture; selectivity is positive there; a deliberately over-capacity probe on few samples shows control CPVI materially above zero (the check detects what it is meant to detect).

Determinism: control labels are seed-reproducible.

Out of scope. Changing the default probe family; the selection rule is a pre-registration decision.

Definition of done. Merged with tests; selectivity present in the RQ1 artefacts; protocol entry written.

DSE-044 — Repeated cross-fit score stabilisation and length control

Epic: E4 · Type: feat · Priority: P1 · Effort: M (5-6h) · Phase: 2 · Dependencies: DSE-014, DSE-028

Context. Per-handoff CPVI from a single cross-fit carries probe-fit noise that can flip instance signs; those scores feed the H2 mediation and all of RQ2. Separately, 'CPVI is just message length' is the first sceptical question and C1 manipulates length directly, so the control belongs in the pre-registration rather than in a rebuttal.

Acceptance criteria. cpvi gains an n_repeats parameter (default 1 to preserve current behaviour; pre-registered value set at freeze) averaging per-instance scores over repeated cross-fits with different fold seeds.

The across-repeat standard deviation per handoff is returned and persisted in scores.parquet as cpvi_sd.

analysis/stats.py gains partial_spearman(x, y, control) and analyse_rq1 reports the partial Spearman of CPVI with outcome given message token length.

The episode-level mediation gains message length as a covariate on path b, reported alongside the uncontrolled version.

Technical notes. Token length derives from message_delivered; no schema change is needed.

Testing requirements. Unit: repeated cross-fits reduce score variance on a noisy fixture; partial Spearman matches a hand-computed value; the length-covariate path fits and is reported.

Out of scope. Choosing R; that is pre-registered from the pilot.

Definition of done. Merged with tests; cpvi_sd persisted; protocol entries added for both.

DSE-045 — Gate retry-feedback template and the greedy fixed-point escape

Epic: E5 · Type: feat · Priority: P0 · Effort: S (3-4h) · Phase: 5 · Dependencies: DSE-017; blocks DSE-018

Context. Under greedy decoding a re-prompt is a fixed point: the same prompt yields the same message, the same statistic and the same block, for all bounded retries. DSE-018 as written would pass its unit tests and be vacuous live. The retry prompt must differ, which makes the feedback template part of the treatment and therefore a pre-registration item, not an implementation detail.

Acceptance criteria. A versioned GATE_FEEDBACK template in agents/prompts.py, appended to A's prompt on a blocked retry, instructing A to state the push direction, whether rotation is needed and which way, and the goal direction explicitly.

GATE_FEEDBACK_VERSION recorded in the run manifest alongside PROMPT_VERSION.

A test proving that a blocked retry issues a different prompt string from the original.

A documented rejection of the nonzero-temperature-on-retry alternative, on the grounds that it breaks the determinism story mid-episode.

Technical notes. The record already carries gate_blocked, gate_retries and message_blocked from the schema v2 bump, so no schema change is needed.

Testing requirements. Unit: the retry prompt differs; the feedback version reaches the manifest; retries are bounded and recorded.

Out of scope. The gate itself (DSE-018) and the RQ3b sweep (DSE-025).

Definition of done. Template merged, versioned and manifested; pre-registration entry written.

DSE-046 — Absent-versus-unused decomposition (the SocialJax replacement)

Epic: E7 · Type: analysis · Priority: P1 · Effort: S (3-4h) · Phase: 3 · Dependencies: DSE-020

Context. Eccles et al. (2019) separate two failures a single correlational number conflates: a sender that fails to encode (positive signalling) and a receiver that fails to act (positive listening). CPVI can tell them apart, which converts a possible RQ1 null into a reportable finding. This delivers the contrast the SocialJax arm was meant to provide, on data you already have, for a day of analysis and no new compute.

Acceptance criteria. analyse_rq1 emits a two-by-two of handoffs split on CPVI (above/below the within-condition median) and realised progress (positive/negative), per condition.

Two named quantities reported with intervals: the absent-signal rate (low CPVI) and the unused-signal rate (high CPVI, poor outcome).

A figure showing how the two rates move across C0 to C4.

A protocol entry framing the decomposition as a pre-registered secondary analysis.

Technical notes. Median-split within condition, not pooled, so the split is not itself a condition effect. The threshold rule is pre-registered.

Testing requirements. Unit: the decomposition reproduces on a fixture with known absent and unused populations.

Out of scope. Any MARL comparison.

Definition of done. Merged with tests; figure and protocol entry produced.

DSE-047 — Re-anchor the RQ3a baselines and the contribution framing

Epic: E7 · Type: docs · Priority: P0 · Effort: S (2-3h) · Phase: 6 · Dependencies: none

Context. The lit review positions H5 against Who&When's 53.5 per cent agent and 14.2 per cent step accuracy. Both have been beaten: AgenTracer-8B reports roughly 69 per cent agent and 20 per cent step on Who&When, and TraceElephant's agentic methods reach 66.7 per cent agent and 33.3 per cent step on full traces. Writing to the old numbers is a viva liability.

Acceptance criteria. A baselines table in the RQ3a design note listing every method, its reported numbers, its substrate and its date.

A revised contribution statement framing CPVI on the prospective-versus-retrospective and cost axes rather than on raw localisation accuracy.

The lit review and abstract updated to match, with the change logged in docs/experiment_design_log.md.

Technical notes. No code. This is a research-writing ticket, deliberately ticketed so it does not live only in a chat.

Testing requirements. None. Human review against the cited sources.

Out of scope. Running any comparison.

Definition of done. Table and framing committed; design-log entry dated.

DSE-048 — Stretch: RuntimeGate against a real MAS via the TraceElephant middleware

Epic: E5 · Type: experiment · Priority: P2 · Effort: XL · Phase: post-RQ1 · Dependencies: DSE-018, DSE-041

Context. TraceElephant ships a non-intrusive LLM API middleware that intercepts and can modify LLM requests and responses without changing the original agent code, plus runnable Captain-Agent, Magentic-One and SWE-Agent systems. That is exactly the seam RuntimeGate needs. If taken, RQ3b's causal claim moves from a 2D simulator to a real multi-agent system, which is a materially stronger result.

Acceptance criteria. A gate adapter behind the middleware that scores the outgoing message with the calibrated Statistic and blocks or passes before the receiver acts.

The same matched-firing-rate and random-trigger controls as the simulator arm.

A small run on one system and one task family, with success, firing rate and retry counts reported against both controls.

Technical notes. Do not attempt this before RQ1 and RQ2 are frozen. It is upside, and it competes for the same time as the write-up.

The threshold must be recalibrated on this domain; transporting the simulator threshold is not defensible.

Testing requirements. Integration against a mocked middleware; the live run is a human-gated experiment.

Out of scope. Full-scale evaluation across all three systems.

Definition of done. Adapter merged with mocked tests; one small live run reported, or the ticket formally deferred with a written reason.

10. Sources

All accessed 25 July 2026. Numbers quoted in this document come from these sources; where a figure is reported in a secondary summary rather than read from the paper's own table it is marked approximate in the text.

Source	Reference and link
Who&When	Zhang, S., Yin, M., Zhang, J., Liu, J., Han, Z., Zhang, J., Li, B., Wang, C., Wang, H., Chen, Y. and Wu, Q. (2025) 'Which Agent Causes Task Failures and When? On Automated Failure Attribution of LLM Multi-Agent Systems', ICML 2025 (Spotlight). arXiv:2505.00212. https://arxiv.org/abs/2505.00212 · Code https://github.com/ag2ai/Agents_Failure_Attribution · Data https://huggingface.co/datasets/Kevin355/Who_and_When
TraceElephant	Chen, M., Wang, J., Mu, F., Wang, Y., Liu, Z., Feng, H. and Wang, Q. (2026) 'Seeing the Whole Elephant: A Benchmark for Failure Attribution in LLM-based Multi-Agent Systems', ACL 2026. arXiv:2604.22708. https://arxiv.org/abs/2604.22708 · Code https://github.com/TraceElephant/TraceElephant · Data https://huggingface.co/datasets/TraceElephant/TraceElephant
AgenTracer	Zhang, G., Wang, J., Chen, J., Zhou, W., Wang, K. and Yan, S. (2025) 'AgenTracer: Who Is Inducing Failure in the LLM Agentic Systems?'. arXiv:2509.03312. https://arxiv.org/abs/2509.03312
MAST	Cemri, M., Pan, M.Z., Yang, S. et al. (2025) 'Why Do Multi-Agent LLM Systems Fail?', NeurIPS 2025 Datasets and Benchmarks Track. arXiv:2503.13657. https://arxiv.org/abs/2503.13657 · Data https://huggingface.co/datasets/mcemri/MAST-Data · Code https://github.com/multi-agent-systems-failure-taxonomy/MAST
TRAIL	'TRAIL: Trace Reasoning and Agentic Issue Localization' (2025), Patronus AI. arXiv:2505.08638. https://arxiv.org/abs/2505.08638 · Code https://github.com/patronus-ai/trail-benchmark · Data https://huggingface.co/datasets/PatronusAI/TRAIL. [Author list to be confirmed from the arXiv listing.]
TELBench / DRIFT	'Where Do Deep-Research Agents Go Wrong? Span-Level Error Localization in Agent Trajectories' (2026). arXiv:2606.02060. https://arxiv.org/abs/2606.02060
SocialJax	Guo, Z., Shi, S., Willis, R., Tomilin, T., Leibo, J.Z. and Du, Y. (2026) 'SocialJax: An Evaluation Suite for Multi-agent Reinforcement Learning in Sequential Social Dilemmas', ICLR 2026. arXiv:2503.14576. https://arxiv.org/abs/2503.14576 · Code https://github.com/cooperativex/SocialJax
Control tasks	Hewitt, J. and Liang, P. (2019) 'Designing and Interpreting Probes with Control Tasks', EMNLP-IJCNLP, pp. 2733-2743. arXiv:1909.03368. https://aclanthology.org/D19-1275/
Conditional probing	Hewitt, J., Ethayarajh, K., Liang, P. and Manning, C.D. (2021) 'Conditional probing: measuring usable information beyond a baseline', EMNLP. arXiv:2109.09234.
PVI	Ethayarajh, K., Choi, Y. and Swayamdipta, S. (2022) 'Understanding dataset difficulty with V-usable information', ICML, PMLR 162. Code https://github.com/kawine/dataset_difficulty
V-information	Xu, Y., Zhao, S., Song, J., Stewart, R. and Ermon, S. (2020) 'A theory of usable information under computational constraints', ICLR. Code https://github.com/Newbeeer/V-information
In-context PVI	Lu, S. et al. (2023) 'Measuring Pointwise V-Usable Information In-Context-ly', Findings of EMNLP. Code https://github.com/UKPLab/in-context-pvi
Emergent comms critique	Lowe, R., Foerster, J., Boureau, Y.-L., Pineau, J. and Dauphin, Y. (2019) 'On the pitfalls of measuring emergent communication', AAMAS. arXiv:1903.05168.
Signalling / listening	Eccles, T., Bachrach, Y., Lever, G., Lazaridou, A. and Graepel, T. (2019) 'Biases for emergent communication in multi-agent reinforcement learning', NeurIPS 32. arXiv:1912.05676.

Counterfactual influence	Jaques, N., Lazaridou, A., Hughes, E., Gulcehre, C., Ortega, P.A., Strouse, D., Leibo, J.Z. and de Freitas, N. (2019) 'Social Influence as Intrinsic Motivation for Multi-Agent Deep Reinforcement Learning', ICML. arXiv:1810.08647.
IMAC	Wang, R., He, X., Yu, R., Qiu, W., An, B. and Rabinovich, Z. (2020) 'Learning efficient multi-agent communication: an information bottleneck approach', ICML, PMLR 119. arXiv:1911.06992.

A note on what I could not verify. AgenTracer's exact agent-level and step-level accuracies (approximately 69 per cent and 20 per cent) come from secondary summaries rather than from the paper's own result table, which I did not read in full; the paper's abstract states the 18.18 per cent relative margin over Gemini-2.5-Pro and Claude-4-Sonnet directly. The claim that MAST-Data contains a usable non-failure class rests on all-zero annotation vectors visible in the HuggingFace preview, not on a counted proportion. Both should be checked in the loader spike before either is cited in the thesis.